

Curriculum Vitae, M. Haque

December 2012

Masudul Haque
Max-Planck Institute for Physics of
Complex Systems (MPI-PKS)
Nöthnitzer Strasse 38
01187 Dresden, Germany.

haque@pks.mpg.de
Tel. office +49-351-871-1113
Tel. mobile +31-6-1116-0834
<http://www.pks.mpg.de/~haque/>

Employment & Education

- **Current institute:** (since October 2006)
Max-Planck Institute for Physics of Complex Systems (MPI-PKS), Dresden, Germany.
Currently Staff Scientist (fixed-term) at MPI-PKS.
Former position title: *Distinguished PKS Postdoctoral Fellow*.
Held one of three such fellowships at MPI-PKS, until September 2009.
- Nov. 2003 to Sept. 2006: Postdoctoral researcher, Utrecht University, the Netherlands. Employed in cold-atom theory group led by H. T. C. Stoof.
- Ph.D., October 2003. Rutgers, State University of New Jersey.
Advisor: Andrei E. Ruckenstein.
Thesis title: *Interaction Effects in the Weakly Repulsive Bose Gas*.
- Graduate student and teaching assistant, 1997 to 2003.
Rutgers, State University of New Jersey; Department of Physics & Astronomy.
- Lecturer, 1996-97. Physics Department, Shahjalal University, Bangladesh.
- M.Sc., 1996; B.Sc., 1995. Shahjalal University, Bangladesh.

Research Interests

- **Area:** Condensed Matter Theory.
Collective phenomena in electronic, nanoscale and cold-atom systems.
- **Specific topics of published research:**
Non-equilibrium phenomena; collective quantum dynamics; quenches & ramps.
Trapped atomic fermions and bosons — polarized fermion pairing; Feshbach resonances; BCS-BEC crossover; vortex dynamics in Bose condensates.
Cross-discipline application of concepts from *Quantum Information Theory* (*e.g.*, entanglement measures) to condensed matter issues.
Topological order and unconventional states in many-particle systems.
Fractional quantum Hall states. Frustrated magnets. Spin chains.
Wigner crystallization. Pomeranchuk instabilities.
Laser-excited semiconductors; excitons. Nonlinear dynamics.

Research Supervision

Citations refer to publication list.

- **Publication Record of Supervision:** Supervised or co-supervised junior researchers for papers [1–3, 5, 13–17, 19–21, 24, 25, 38–40].
- **PhD students and postdocs:**
Oleksandr Zozulya. Former PhD student of K. Schoutens at University of Amsterdam. I supervised his PhD research closely; all his PhD publications are coauthored with me [14, 15, 20, 24, 25].
Wladimir Tschischik. PhD student with me at MPI-PKS since April 2012. Pursuing research on non-equilibrium dynamics in Bose-Hubbard systems ([1]+ongoing). (R. Moessner, the Department Director, is the formal supervisor.)
Shreyoshi Mondal. Postdoc under my supervision at MPI-PKS (Sep. 2012 – present). Working on non-equilibrium dynamics of the Kondo cloud.
Wouter Beugeling. Postdoc under my supervision at MPI-PKS (Oct. 2012 – present). Working on the Eigenstate Thermalization Hypothesis (non-equilibrium physics).
Vincenzo Alba. Postdoc at MPI-PKS (Oct. 2010 – Sep. 2012). Joint supervision with A. M. Läuchli. Research on the nature of entanglement spectra in 1D and 2D systems ([2, 5]+ongoing).

Weibin Li. Former postdoc at MPI-PKS. Worked with me on vortex dynamics [21].

Alexey Mikaberidze. PhD student at MPI-PKS until 2011. Worked with me on an ecology-motivated calculation [19]. Based on this work, Mikaberidze moved to a career in theoretical ecology after his PhD in atomic/laser physics.

Frank Zimmer. Former postdoc at MPI-PKS (2009-2011). Worked with me on non-equilibrium dynamics (non-adiabatic ramps in trapped gases) [39, 40].

V. Ravi Chandra & J. Bandyopadhyay. Former postdocs at MPI-PKS. Worked with me on entanglement and dynamics in frustrated spin chains [17].

Ivana Vidanovic. During PhD studies (in Belgrade), worked with me on non-equilibrium dynamics in spinor condensates [38].

Kush Saha. PhD student in India. Visited MPI-PKS Sep.–Nov. 2012 to work with me. Explored the Bethe ansatz description of some eigenstates of the open Heisenberg chain.

Ricardo Pinto. Former PhD student at MPI-PKS. Worked with me on non-equilibrium dynamics in 1D lattice models (interaction-induced edge effects) [16].

- **Undergraduate Research Supervision (longer project):**

April 2011 – March 2012: Wladimir Tschischik, student from Technische Universität Chemnitz, performed one-year ‘Diplomarbeit’ research with me at MPI-PKS. Defended ‘Diplomarbeit’ thesis in March 2012. Research on non-equilibrium dynamics in Bose-Hubbard ladders [1]. Tschischik is continuing his research with me as PhD student.

- **Undergraduate Research Supervision (short projects):**

Supervising summer research projects for visiting undergraduates.

Summer 2009: T. Venumadhav, visited Dresden from I.I.T. (Kanpur, India); 10 weeks. Research on non-equilibrium dynamics, led to publication; paper [13]. Venumadhav is now pursuing doctoral studies at Caltech.

February–July 2009: S. Piatecki, visited Dresden from ENS Paris; 5 months. Research on geometrically frustrated itinerant systems, later completed with additional collaborators [3, 7]. Piatecki is now pursuing doctoral studies at ENS Paris.

Summer 2010: A. Balram, visited Dresden from ISERC (Pune, India); 3 months. Research on non-equilibrium dynamics in Bose-Hubbard rings. Balram is now pursuing doctoral studies at Penn State University (Pittsburgh).

Summer 2011: S. Chatterjee, visited Dresden from I.I.T. (Kanpur, India); 10 weeks. Research on non-equilibrium dynamics; continued Balram’s work. Chatterjee is starting doctoral studies at Harvard.

Organization activities

Workshop in June 2010

Title: Quantum information concepts for condensed matter problems (QICCMP)
Co-organizers: Ian Affleck (UBC, Canada); Ulrich Schollwöck (LMU, Munich).
Format: Two-week programme (June 14–25, 2010). 100⁺ participants.
Location & Funding: Max Planck Institute (MPI-PKS), Dresden, Germany.
Website: <http://www.pks.mpg.de/~qiccmp10/>

I conceived this event, assembled the organization team, and handled most details as the main organizer and contact person.

Workshop in November 2012

Title: Entanglement Spectra in Complex Quantum Wavefunctions (ESiCQW)
Co-organizers: B. Andrei Bernevig (Princeton); Andreas Läuchli (Innsbruck).
Format: One-week high-intensity workshop (November 12–16, 2012).
Location & Funding: Max Planck Institute (MPI-PKS), Dresden, Germany.
Website: <http://www.pks.mpg.de/~esicqw12/>

Workshop in August 2013

Title: Quantum many body systems out of equilibrium (QSOE)
Co-organizers: J. S. Caux (Amsterdam); C. Kollath (Geneva); T. Esslinger (Zürich).
Format: Three-week programme (August 12–30, 2013).
Location & Funding: Max Planck Institute (MPI-PKS), Dresden, Germany.
Website: <http://www.pks.mpg.de/~qsoe13/>

Other initiatives

Organizing various lectures, talks, journal clubs, locally, *e.g.*,

- a journal club on non-equilibrium physics (quenches & ramps), May 2012 – present.
- a journal club on cold-atom experiments, Oct. 2011 – present.
- a discussion sequence & journal club on interacting fractional Chern insulators, 2011.
- a series of talks on “Cold atoms & BEC”, 2008–2010.
- a lecture series on “1D & Exact Integrability” in Utrecht, spring 2006.
- the condensed matter theory seminar sequence at Utrecht, 2004–2006.

..... *etc.*

Publication List

• Overview & highlights:

- Citations: 700⁺, in November 2012.
For selected papers (15⁺ citations), individual counts (November 2012) are noted below.
Most cited: papers 27, 28, 25, 24, 10.
Citation data from <http://adsabs.harvard.edu/>
- Single-author publications: 9, 18, 29.
- Articles in higher-profile journals:
In *Physical Review Letters*: 2, 6, 7, 10, 25, 26, 27, 31.
In *New Journal of Physics*: 11.
- Paper 14 is a review article.
- Articles 8 and 27 are coauthored with experimental groups.
- Articles highlighted with ▲ were coauthored solely with junior scientists or peers, supervised or led by myself.

• Published or Accepted:

1. *Non-equilibrium dynamics in Bose-Hubbard ladders.*
W. Tschischik, M. Haque, and R. Moessner; Phys. Rev. A, accepted (2012).
Available as preprint: arXiv:1209.5534.
2. *Boundary-locality and perturbative structure of entanglement spectra in gapped systems.*
V. Alba, M. Haque, and A. M. Läuchli; Phys. Rev. Lett. **108**, 227201 (2012).
3. *Itinerant electrons in the Coulomb phase.*
L. D. C. Jaubert, S. Piatecki, M. Haque, and R. Moessner;
Phys. Rev. B **85**, 054425 (2012).
4. *Strongly interacting one-dimensional bosons in optical lattices of arbitrary depth: ▲
from the Bose-Hubbard to the sine-Gordon regime and beyond.*
A. Lazarides and M. Haque, Phys. Rev. A **85**, 063621 (2012).
5. *Entanglement spectrum of the Heisenberg XXZ chain near the ferromagnetic point.*
V. Alba, M. Haque, and A. M. Läuchli; J. Stat. Mech. P08011 (2012).
6. *Crossover from adiabatic to sudden interaction quench in a Luttinger liquid.*
B. Dóra, M. Haque, and G. Zaránd; Phys. Rev. Lett. **106**, 156406 (2011).
★ 15⁺ citations.
7. *Analysis of a fully packed loop model arising in a magnetic Coulomb phase.*
L. D. C. Jaubert, M. Haque, and R. Moessner; Phys. Rev. Lett. **107**, 177202 (2011).

8. *Three-vortex configurations in trapped Bose-Einstein condensates.*
J. A. Seman, E. A. L. Henn, M. Haque, R. F. Shiozaki, E. R. F. Ramos,
M. Caracanhas, P. Castilho, C. Castelo Branco, P. E. S. Tavares,
F. J. Poveda Cuevas, G. Roati, K. Magalhaes, and V. S. Bagnato;
Phys. Rev. A, **82**, 033616 (2010).
★ 15⁺ citations.
9. *Self-similar spectral structures and edge-locking hierarchy in open-boundary spin chains.* — M. Haque, Phys. Rev. A **82**, 012108 (2010).
10. *Disentangling Entanglement Spectra of Fractional Quantum Hall States on Torus Geometries.* — A. M. Laeuchli, E. J. Bergholtz, J. Suorsa, and M. Haque;
Phys. Rev. Lett. **104**, 156404 (2010).
★ 40⁺ citations.
11. *Entanglement Scaling of Fractional Quantum Hall states through Geometric Deformations.* — A. M. Laeuchli, E. J. Bergholtz, and M. Haque;
New Journal of Physics **12**, 075004 (2010).
12. *Interaction induced fractional Bloch and tunneling oscillations.*
R. Khomeriki, D. O. Krimer, M. Haque, and S. Flach;
Phys. Rev. A **81**, 065601 (2010).
13. *Finite-rate quenches of site bias in the Bose-Hubbard dimer.*
T. Venumadhav, M. Haque, and R. Moessner; Phys. Rev. B **81**, 054305 (2010).
14. *Entanglement between particle partitions in itinerant many-particle states.*
M. Haque, O. S. Zozulya, and K. Schoutens;
J. Phys. A: Math. Theor. **42**, 504012 (2009).
15. *Entanglement signatures of Quantum Hall phase transitions.* ▲
O. Zozulya, M. Haque, and N. Regnault; Phys. Rev. B **79**, 045409 (2009).
★ 20⁺ citations.
16. *Edge-localized states in quantum one-dimensional lattices.*
R. A. Pinto, M. Haque, and S. Flach; Phys. Rev. A **79**, 052118 (2009).
17. *Entanglement and level crossings in frustrated ferromagnetic rings.* ▲
M. Haque, J. N. Bandyopadhyay, and V. Ravi Chandra;
Phys. Rev. A **79**, 042317 (2009).
18. *Probing topological order in quantum Hall states using entanglement calculations.*
M. Haque; AMS Contemp. Math. **482**, 213 (2009).
19. *Survival benefits in mimicry: a quantitative framework.* ▲
A. Mikaberidze and M. Haque; J. Theor. Biol. **259**, 462 (2009).
20. *Particle partitioning entanglement in itinerant many-particle systems.*
O. Zozulya, M. Haque, and K. Schoutens; Phys. Rev. A **78**, 042326 (2008).
21. *A vortex dipole in a trapped two-dimensional Bose condensate.* ▲
W. Li, M. Haque and S. Komineas; Phys. Rev. A **77**, 053610 (2008).
★ 20 citations.

22. *Symmetry-breaking Fermi surface deformations from central interactions in two dimensions.*
J. Quintanilla, M. Haque, and A. J. Schofield; Phys. Rev. B, **78**, 035131 (2008).
23. *Pomeranchuk instability: symmetry breaking and experimental signatures.*
J. Quintanilla, C. Hooley, B. J. Powell, A. J. Schofield, and M. Haque;
Physica B: Condensed Matter, **403**, 1279 (2008).
24. *Bipartite entanglement entropy in fractional quantum Hall states.*
O. Zozulya, M. Haque, K. Schoutens, and E. H. Rezayi;
Phys. Rev. B **76**, 125310 (2007).
★ 40⁺ citations.
25. *Entanglement entropy in fermionic Laughlin states.*
M. Haque, O. Zozulya, and K. Schoutens; Phys. Rev. Lett. **98**, 060401 (2007).
★ 60⁺ citations.
26. *Trapped fermionic clouds distorted from the trap shape due to many-body effects.*
M. Haque and H. T. C. Stoof; Phys. Rev. Lett., **98**, 260406 (2007).
★ 20⁺ citations.
27. *Deformation of a Trapped Fermi Gas with Unequal Spin Populations.*
G. B. Partridge, Wenhui Li, Y. A. Liao, R. G. Hulet, M. Haque, and H. T. C. Stoof;
Phys. Rev. Lett. **97**, 190407 (2006).
★ 180⁺ citations.
28. *Pairing of a trapped resonantly interacting fermion mixture with unequal spin populations.* — M. Haque and H. T. C. Stoof; Phys. Rev. A **74**, 011602 (2006).
★ 90⁺ citations.
29. *Ring-shaped luminescence pattern in biased quantum wells studied as a steady state reaction front.* — M. Haque; Phys. Rev. E **73**, 066207 (2006).
30. *Squeezing in the weakly interacting uniform Bose-Einstein condensate.*
M. Haque and A. E. Ruckenstein; Phys. Rev. A, **74**, 043622 (2006).
31. *Ultracold superstring in atomic boson-fermion mixtures.*
M. Snoek, M. Haque, S. Vandoren, and H. T. C. Stoof;
Phys. Rev. Lett. **95**, 250401 (2005).
★ 20⁺ citations.
32. *Dynamics of a molecular Bose-Einstein condensate near a Feshbach resonance.*
M. Haque and H. T. C. Stoof; Phys. Rev. A **71**, 063603 (2005).
33. *Structural Transition of Wigner Crystal on Liquid Substrate.* ▲
M. Haque, I. Paul and S. Pankov; Phys. Rev. B **68**, 045427 (2003).
34. *Anomaly in the normalization constant of the (³He,p) reaction.*
M. Haque *et. al.*, Nuovo Cimento A, **111A**, 1131 (1998).
35. *⁶⁴Cu levels from the ⁶²Ni(³He,p) reaction at 18 MeV.*
A. K. Basak *et. al.*, Phys. Rev. C **56**, 1983 (1997).

- **Publicly available preprints (under review or unpublished):**

36. *Strongly interacting bosons in multi-chromatic potentials supporting mobility edges: localization, quasi-condensation and expansion dynamics.* ▲
P. Ribeiro, M. Haque, and A. Lazarides; arXiv:1211.6012.
37. *Linear quantum quench in the Heisenberg XXZ chain: time dependent Luttinger model description of a lattice system.*
F. Pollmann, M. Haque, and B. Dora; arXiv:1211.1195.
38. *Spin modulation instabilities and phase separation dynamics in trapped two-component Bose condensates.*
I. Vidanovic, N. J. Van Druten, and M. Haque, arXiv:1210.0030.
39. *Slow interaction ramps in trapped many-particle systems: universal deviations from adiabaticity.* — M. Haque and F. E. Zimmer, arXiv:1110.0840. ▲
40. *Non-adiabatic interaction ramps in a trapped Bose condensate.* ▲
F. E. Zimmer and M. Haque, arXiv:1012.4492.
41. *Transition Temperature of Dilute Weakly Repulsive Bose Gas.*
M. Haque and A. E. Ruckenstein, cond-mat/0212590.
42. *Weakly Non-ideal Bose Gas: Comments on Critical Temperature Calculations.*
M. Haque, cond-mat/0302076.